

Supply Chain Management

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Ch 12. Managing Uncertainty in a Supply Chain: Safety Inventory



Chapter Overview

1. The Role of Safety Inventory in a Supply Chain
2. Factors Affecting the Level of Safety Inventory
3. Determining the Appropriate Level of Safety Inventory
4. Impact of Supply Uncertainty on Safety Inventory
5. Impact of Aggregation on Safety Inventory
6. Impact of Replenishment Policies on Safety Inventory
7. Managing Safety Inventory in a Multi-echelon Supply Chain
8. Managerial Levers to Reduce Safety Inventory



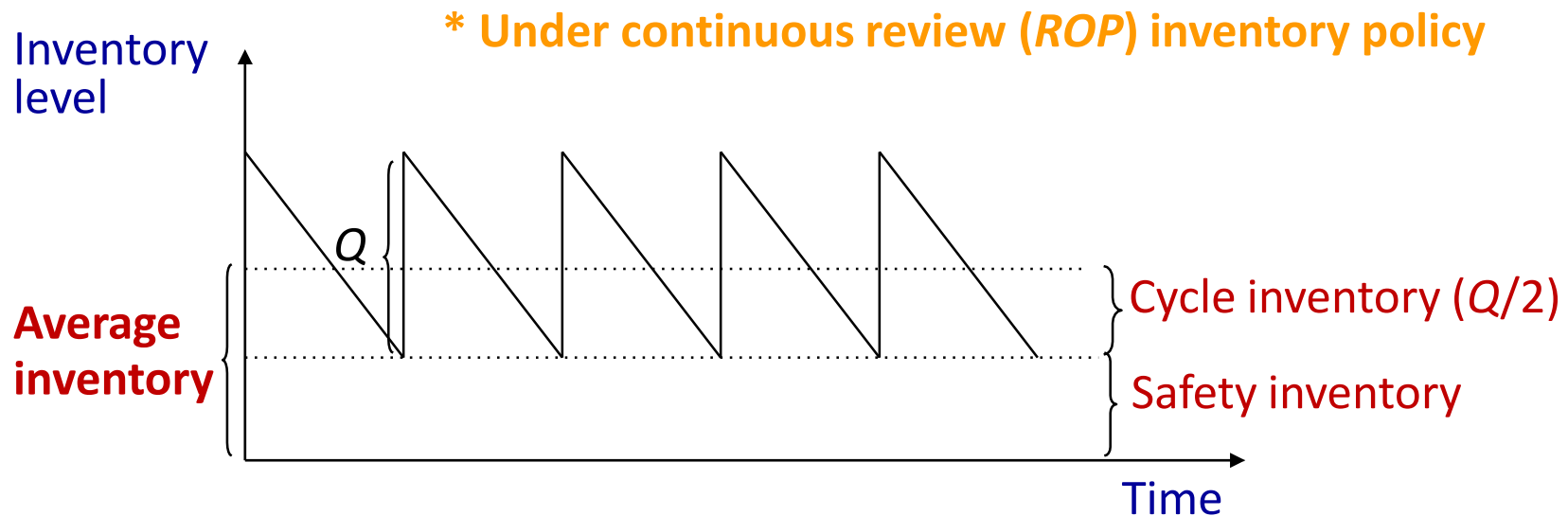
12.1. The Role of Safety Inventory in a Supply Chain

- Demand and lead time are uncertain
- **Safety inventory(stock)** carried for the purpose of satisfying demand that exceeds the amount forecasted (i.e., over average demand) in a given period
- More safety inventory due to increased product variety, shortened product life, and globalization
- There is a **fundamental trade-off** between the level of product availability (or customer service) and inventory holding costs
- Especially, the reduction in safety inventory is very important in high-tech industries
(e.g.) LED TV, semi-conductor, cell phones, sensors, aircraft parts



12.1. The Role of Safety Inventory in a Supply Chain

- Three Key Questions to Answer When Planning Safety Inventory
 - 1) What is the appropriate level of product availability? (Chap. 13)
 - 2) How much safety inventory is needed for the desired level of product availability?
 - 3) What actions can be taken to reduce safety inventory without hurting product availability?



12.2. Factors Affecting the Level of Safety Inventory

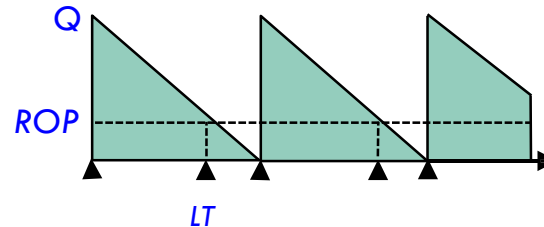
- Appropriate level of safety inventory determined by:
 - Desired level of product availability
 - Uncertainty of demand
 - Uncertainty of supply (or lead time)
 - Inventory replenishment policies
- Higher level of supply or demand uncertainty requires higher level of safety inventory given a particular desired level of product availability
- Higher level of desired product availability requires higher level of safety inventory given a particular level of uncertainty



12.2. Factors Affecting the Level of Safety Inventory

- Measuring product availability

- Product availability reflects a firm's ability to fill a customer order out of available inventory



- Three measures

- 1) **Cycle service level (CSL)**: Probability of not having a stockout in a replenishment cycle, that is, fraction of replenishment cycles that end with all customer demand being met

$$CSL = Prob(\text{demand during lead time} \leq ROP)$$

- 2) **Product fill rate (fr)**: Fraction of demand satisfied from product in inventory

$$fr = 1 - ESC/Q$$

where ESC : expected shortage per replenishment cycle
 Q : also the average demand in a replenishment cycle

- 3) **Order fill rate**: Fraction of orders filled from available inventory



12.2. Factors Affecting the Level of Safety Inventory

- Measuring product availability
 - Demand uncertainty is usually estimated by the standard deviation of demand
 - Demand uncertainty during lead time is important under continuous review (or ROP) replenishment policy
 - Notations
 - D : average demand per period
 - σ_D : standard deviation of demand per period
 - L : lead time (i.e., gap between when an order is placed and when it is received)
 - D_L : average demand during lead time
 - σ_L : standard deviation of demand during lead time



12.2. Factors Affecting the Level of Safety Inventory

- Evaluating demand distribution over L periods
 - When demand for each period $i \sim (D_i, \sigma_i)$ and $\rho_{ij} \neq 0$, the total demand during L periods is normally distributed:

$$D_L = \sum_{i=1}^L D_i \quad \sigma_L = \sqrt{\sum_{i=1}^L \sigma_i^2 + 2 \sum_{i>j} \rho_{ij} \sigma_i \sigma_j}$$

- When demand for each period $i \sim \text{NORM}(D, \sigma_D)$ and $\rho_{ij} = 0$, the total demand during L periods is normally distributed:

$$D_L = L \times D, \sigma_L = \sqrt{L} \times \sigma_D$$

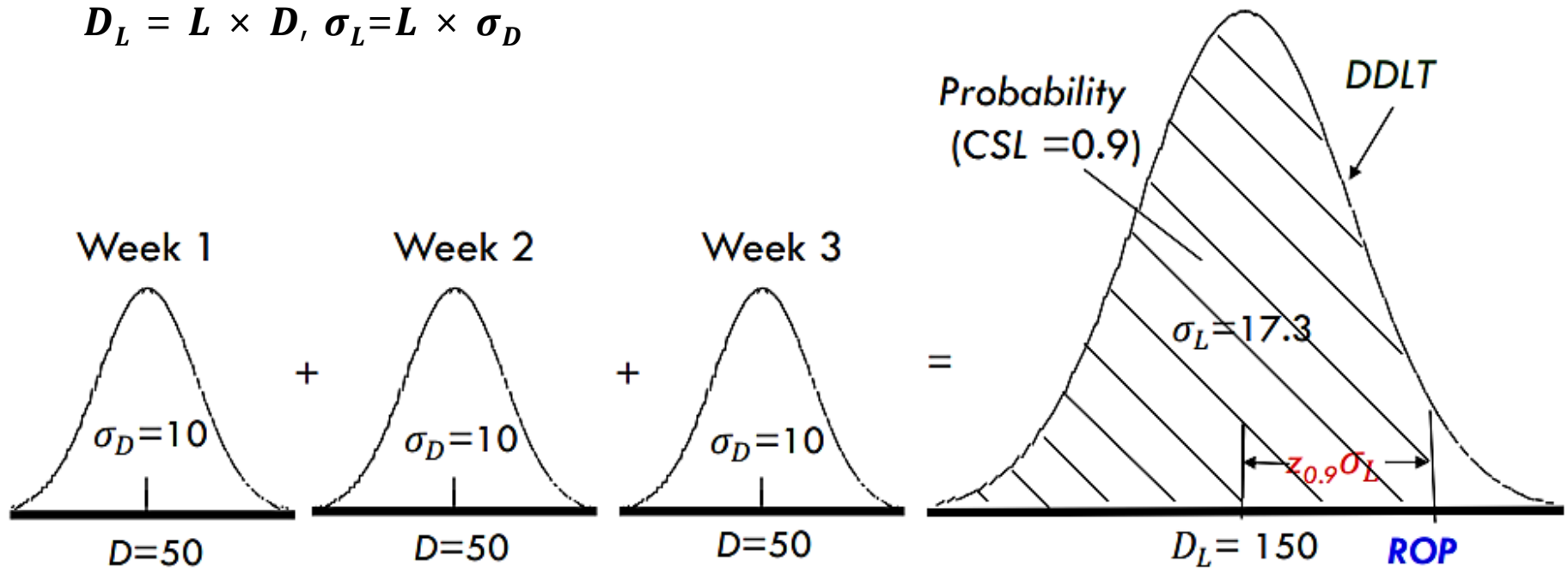
- Another important measure of demand uncertainty
 - σ/μ : Coefficient of variation
(i.e., size of the uncertainty relative to mean)



12.2. Factors Affecting the Level of Safety Inventory

- Example: Demand distribution during $L = 3$ weeks

$$D_L = L \times D, \sigma_L = L \times \sigma_D$$



Weekly demand $\sim \text{NORM}(50, 10)$

Lead time is 3 weeks

$$D_L = L \times D = 3 \times 50 = 150$$

$$\sigma_L = \sqrt{L} \times \sigma_D = \sqrt{3} \times 10 = 17.3$$

$$\begin{aligned} ROP &= D_L + z_{0.9} \times \sigma_L \\ &= 150 + 1.28 \times 17.3 = 172 \end{aligned}$$

$$ss = ROP - D_L = z_{0.9} \times \sigma_L = 22$$



12.3. Determining the Appropriate Level of Safety Inventory

- Given a continuous review policy, two measures of product availability are considered:
 - Cycle service level
 - Product fill rate

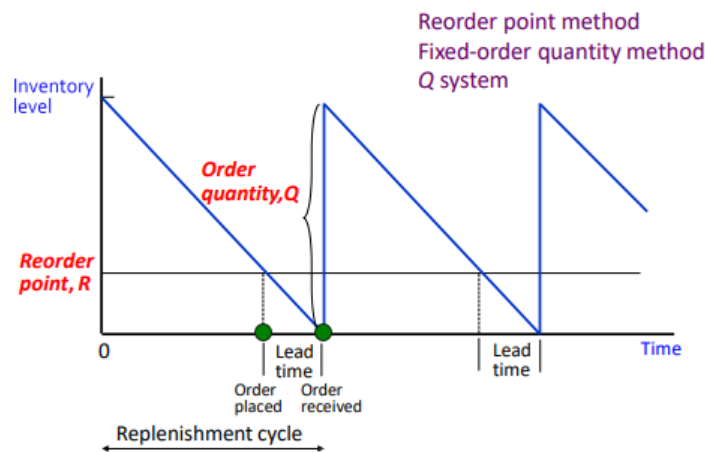


12.3. Determining the Appropriate Level of Safety Inventory

- Replenishment Policies

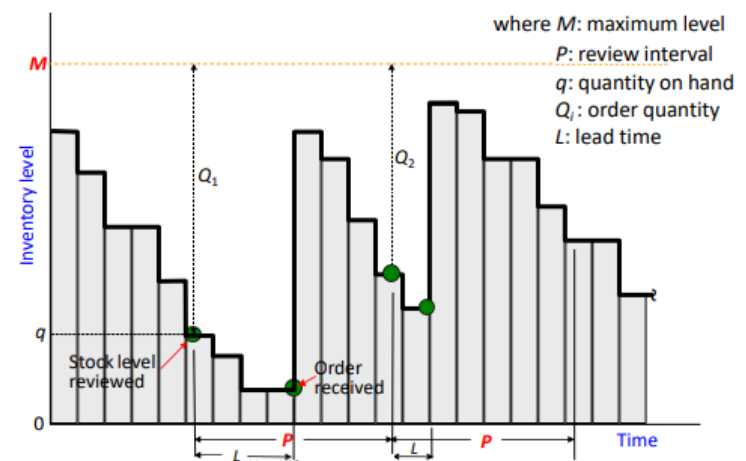
- Replenishment policy consists of decisions regarding **when to reorder** and **how much to reorder** which determine safety inventories and product availability
 - **Continuous review***: Inventory is continuously monitored and an order of size Q is placed when the inventory level reaches the reorder point **ROP**
 - **Periodic review**: Inventory is checked at regular (periodic) intervals and an order is placed to raise the inventory to a specified threshold (i.e., the “**order-up-to**” level)

- Continuous Review Method



- Periodic Review Method

- Fixed-period method, P system



12.3. Determining the Appropriate Level of Safety Inventory

- Cycle service level

Given ROP ,

$CSL = \text{Prob}(\text{Demand during } L \leq ROP)$

$$(1) \text{ } CSL = \int_{-\infty}^{ROP} f(x)dx = F(ROP, D_L, \sigma_L)$$

$$(2) \text{ } CSL = \text{Prob}\left(z \leq \frac{ROP - D_L}{\sigma_L}\right)$$

Given CSL ,

$$(1) \text{ } ss = F_s^{-1}(CSL) \times \sigma_L$$

$$(2) \text{ } ss = z_{CSL} \times \sigma_L \quad (ROP = D_L + ss)$$

(Appendix 12A & 12B)

Notations

L : lead time for replenishment

D : average demand per period

σ_D : standard deviation of demand per period

D_L : mean demand during lead time

σ_L : standard deviation of demand during lead time

ss : safety inventory

ROP : reorder point

$f(x)$: density function of demand distribution during lead time

Using EXCEL functions,

$CSL = \text{NORMDIST}(ROP, D_L, \sigma_L, 1)$

$ss = \text{NORMSINV}(CSL) \times \sigma_L$



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.1: Evaluating safety inventory given a reorder point

- B&M Office Supplies: Phone sale

Weekly $D \sim NORM(2500, 500)$, $L = 2$ weeks, $Q = 10,000$. When $ROP = 6,000$:

- Analysis:

- $D_L = D \times L = (2,500)(2) = 5,000$
 - $ss = ROP - D_L = 6,000 - 5,000 = 1,000$
 - Cycle inventory = $Q/2 = 10,000/2 = 5,000$
 - Average inventory = cycle inventory + $ss = 5,000 + 1,000 = 6,000$
 - Average flow time = average inventory/throughput = $6,000/2,500 = 2.4$ weeks



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.2: Evaluating cycle service level given a reorder point

- B&M Office Supplies: Phone sale

$D = 2,500/\text{week}, \sigma_D = 500, L = 2 \text{ weeks}, Q = 10,000.$

When $ROP = 6,000$, evaluate the safety inventory and CSL

$$D_L = 5,000, \sigma_L = \sigma_D \sqrt{L} = 500\sqrt{2} = 707$$

$$ss = 6,000 - 5,000 = 1,000$$

$$CSL = Prob(\text{Demand during } L \leq ROP)$$

$$= F(ROP, D_L, \sigma_L) = NORMDIST(ROP, D_L, \sigma_L, 1)$$

$$= NORMDIST(6000, 5000, 707, 1) = \mathbf{0.92} \quad \text{or}$$

$$= Prob\left(z \leq \frac{6000-5000}{707}\right) = Prob(z \leq 1.41) = \mathbf{0.92}$$

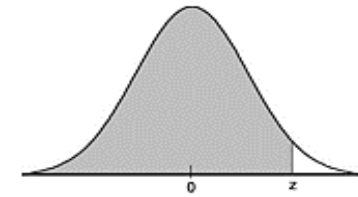
(from a standardized normal distribution table in the next slide)



12.3. Determining the Appropriate Level of Safety Inventory

Areas under Standardized Normal Distribution

A table entry is the proportion of the area under the curve from a z of 0 to a positive value of z . To find the area from a z of 0 to a negative z , subtract the tabled value from 1.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.3: Evaluating safety inventory given a desired cycle service level

- Carrefour Store: Lego Sale

$D = 2,500/\text{week}$, $\sigma_D = 500$, $L = 2$ weeks, $Q = 10,000$

Evaluate the safety inventory that the store should carry to achieve CSL of 90%.

- Analysis:

$$D_L = 5,000, \sigma_L = \sigma_D \sqrt{L} = 500\sqrt{2} = 707$$

$$ss = F_S^{-1}(0.9) \times \sigma_L$$

$$= \text{NORMSINV}(0.9) \times 707 = \mathbf{906} \text{ or}$$

$$= z_{0.9} \times \sigma_L = 1.28 \times 707 = \mathbf{906}$$

(Less than 1,000 in Example 12.2 where $CSL = 92\%$)



12.3. Determining the Appropriate Level of Safety Inventory

- Product fill rate(fr): Proportion of customer demand satisfied from inventory

$$fr = 1 - \frac{ESC}{Q}$$
$$ESC = \int_{x=ROP(=D_L+ss)}^{\infty} (x - ROP) f(x) dx$$
$$ESC = -ss \left\{ 1 - F_s \left(\frac{ss}{\sigma_L} \right) \right\} + \sigma_L f_s \left(\frac{ss}{\sigma_L} \right)$$

(Appendix 12C)

Using EXCEL function:

$$ESC = -ss \{ 1 - NORMDIST(ss/\sigma_L, 0, 1, 1) \} + \sigma_L NORMDIST(ss/\sigma_L, 0, 1, 0)$$

Notations

ESC: expected shortage per cycle
(average demand in excess of reorder point in each replenishment cycle)

Q: order quantity (or average demand) per cycle

$f(x)$: density function of demand distribution during lead time



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.4: Evaluating fill rate given a reorder point

- B&M Office Supplies: Phone sale

$$Q = 10,000, L = 2 \text{ weeks}, D_L = 5,000, \sigma_L = 500\sqrt{2}=707.$$

When ROP = 6,000, evaluate the product fill rate.

- Analysis:

$$ss = 6,000 - 5,000 = 1,000$$

$$\begin{aligned} ESC &= -ss\{1 - \text{NORMDIST}(ss/\sigma_L, 0, 1, 1)\} + \sigma_L \text{NORMDIST}(ss/\sigma_L, 0, 1, 0) \\ &= -1000\{1 - \text{NORMDIST}(1000/707, 0, 1, 1)\} \\ &\quad + 707\text{NORMDIST}(1000/707, 0, 1, 0) = 25 \end{aligned}$$

$$fr = (Q - ESC)/Q = (10,000 - 25)/10,000 = \mathbf{0.9975}$$

**Much higher than *CSL* of 0.92 for the same replenishment policy
being used in Example 12.2 (*Why?*)**



12.3. Determining the Appropriate Level of Safety Inventory

- Factors affecting fill rate

$$fr = 1 - \frac{ESC}{Q}$$

- **Safety inventory:** Fill rate increases if safety inventory is increased (this also increases the cycle service level)
- **Lot size:** Fill rate increases if lot size is increased, but cycle service level does not change

- **Example**

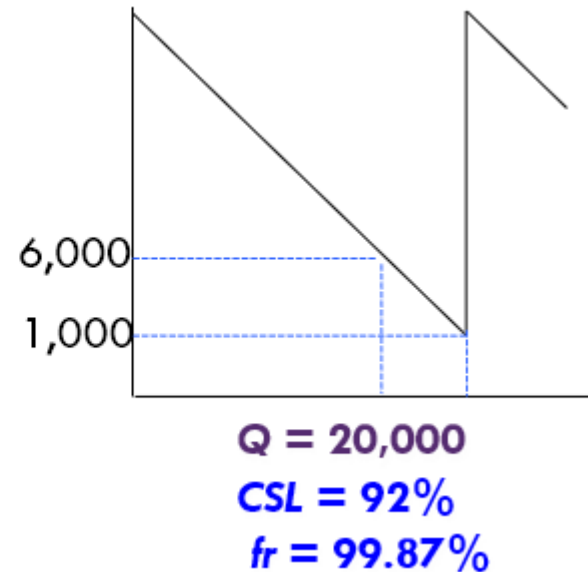
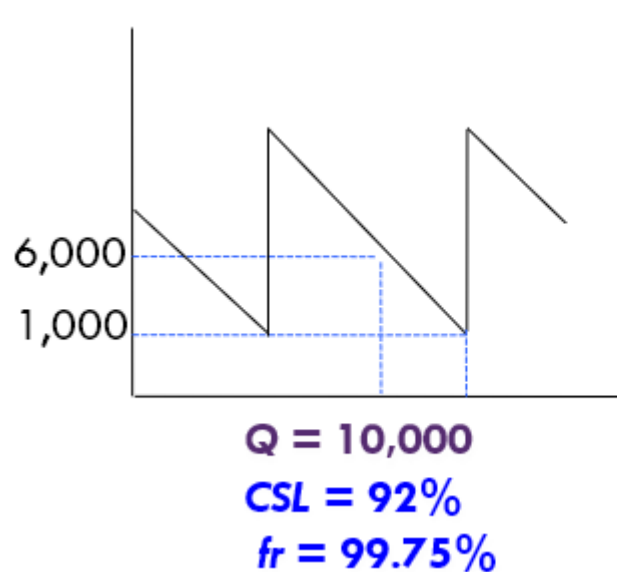
When the lot size is increased from 10,000 to 20,000:

$CSL = 0.92$ (no change), $fr = 0.9975 \rightarrow 0.9987$



12.3. Determining the Appropriate Level of Safety Inventory

- CSL vs. Product fill rate
 - ROP and ss are given as 6,000 and 1,000, each:



- Order quantity is getting **larger**, then the probability that customer order can be satisfied from the available inventory is also getting **bigger**, hence with a **bigger** fr
- CSL considers only the lead time, is not dependent on the Q



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.5: Evaluating safety inventory given desired fill rate

- Carrefour Store: Lego Sale

$D = 2,500/\text{week}$, $\sigma_D = 500$, $Q = 10,000$, $L = 2$ weeks, $\sigma_L = 707$

When $fr = 0.975$, evaluate the safety inventory

- Analysis:

$$ESC = (1 - fr)Q = 250 \quad (\text{from } fr = 1 - ESC/Q)$$

$$ESC = 250 = -ss \left[1 - F_S \left(\frac{ss}{\sigma_L} \right) \right] + \sigma_L f_S \left(\frac{ss}{\sigma_L} \right) \quad \text{Eqs. 12.9 \& 12.10}$$

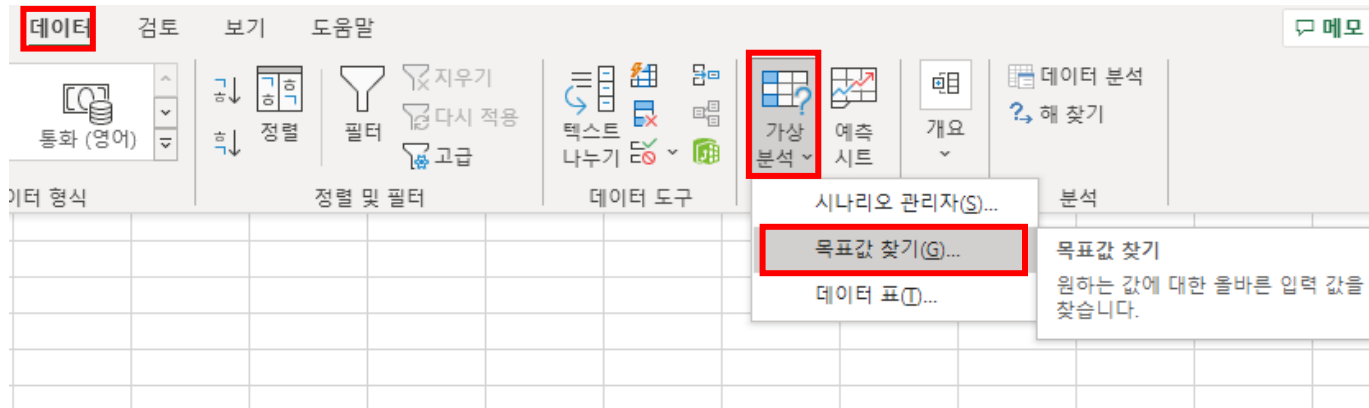
$$250 = -ss \left[1 - NORMSDIST \left(\frac{ss}{\sigma_L}, 0, 1, 1 \right) \right] + \sigma_L NORMDIST \left(\frac{ss}{\sigma_L}, 0, 1, 0 \right)$$

Using the Excel tool GOALSEEK, $ss = 67$ boxes



12.3. Determining the Appropriate Level of Safety Inventory

- Spreadsheet to solve for safety stock using GOALSEEK (pp. 337)



	A	B	C	D	E
1	Input			Calculation	Variable
2	fr	σ_L	Q	Desired ESC	ss
3	0.975	707	10000	250	67
4	Formula				
5	Actual ESC				
6	250				
7					
8					
9					
10					
11					

Goal Seek

Set cell:

To value:

By changing cell:

OK Cancel

Cell	Cell Formula	Equation
A6	$-E3*(1-NORMSDIST(E3/B3, 0, 1, 1)) + B3*NORMDIST(E3/B3, 0, 1, 0)$	12.10



12.3. Determining the Appropriate Level of Safety Inventory

- Impact of desired product availability, lead time, and demand uncertainty on safety inventory
 - Required safety inventory increases as desired product availability (CSL or fr), lead time, and/or demand uncertainty increase
 - $SS = z_{CSL}(\sigma_L) = z_{CSL}(\sqrt{L} \cdot \sigma_D)$
 - Managerial levers to reduce safety inventory without reducing product availability
 - Better relationships with suppliers to reduce the supplier leadtime (The required safety stock decreases by a factor of $\sqrt{k}$)
 - Better forecasts by more sophisticated methods and improved coordination among partners to reduce the uncertainty of demand (The required safety stock decreases by a factor of k)

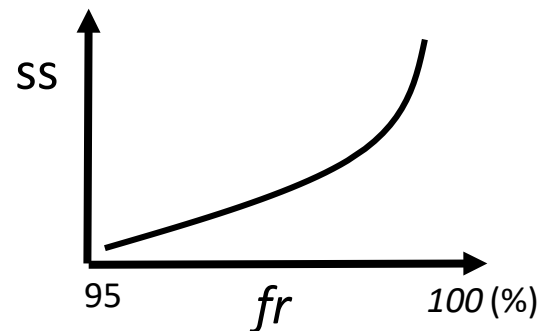


12.3. Determining the Appropriate Level of Safety Inventory

- Required safety inventory for different values of fill rate

Product Fill Rate (fr)	Safety Inventory (boxes)
97.5%	67
98.0%	183
98.5%	321
99.0%	499
99.5%	767

(Using Example 12.5 while fixing $Q = 10,000$)



12.3. Determining the Appropriate Level of Safety Inventory

- Example 12.6: Benefits of reducing lead time and demand uncertainty
 - Target Store: White Shirts Sale
 $D = 2,500/\text{week}$, $\sigma_D = 800$, $L = 9$ weeks, $CSL = 0.95$
 - What savings in safety inventory can the store expect if the supplier reduces lead time to one week? What savings in safety inventory if the standard deviation of demand is reduced to 400?
 - Analysis:

$$\begin{aligned}ss &= NORMSINV(CSL) \times \sqrt{L} \sigma_D \quad \text{or } z_{0.95} \times \sqrt{L} \sigma_D \\&= NORMSINV(.95) \times \sqrt{9} \times 800 = 3,948\end{aligned}$$

- If lead time is reduced to one week

$$ss = NORMSINV(.95) \times \sqrt{1} \times 800 = 1,316$$

- If standard deviation is reduced to 400

$$ss = NORMSINV(.95) \times \sqrt{9} \times 400 = 1,974$$



12.4. Impact of Supply Uncertainty on Safety Inventory

- Supply uncertainty is expressed as **the uncertainty in supply lead time**
- Under a continuous review policy, demand during the lead time is normally distributed as follows:

$$D_L = D \times L$$
$$\sigma_L = \sqrt{L\sigma_D^2 + D^2s_L^2}$$

where D : average demand per period

σ_D : standard deviation of demand per period

L : average lead time

s_L : standard deviation of lead time

- Given CSL , $ss = NORMSINV(CSL) \times \sigma_L$ or $Z_{CSL} \times \sigma_L$



12.4. Impact of Supply Uncertainty on Safety Inventory

- Example 12.7: Impact of lead time uncertainty on safety inventory

- Amazon: Tablet PCs

$D = 2,500/\text{day}$, $\sigma_D = 500$, $L = 7$ days, $Q = 10,000$

Targeting a *CSL* of 90%, evaluate the safety inventory of tablets that Amazon must carry if the **standard deviation of lead time is seven days ($S_L = 7$)**.

- Analysis:

$$D_L = DL = (2500)(7) = 17,500$$

$$\begin{aligned}\sigma_L &= \sqrt{L\sigma_D^2 + D^2S_L^2} \\ &= \sqrt{(7)500^2 + (2500)^2(7)^2} = 17,550\end{aligned}$$

$$\begin{aligned}ss &= F_s^{-1}(\text{CSL}) \times \sigma_L = \text{NORMSINV}(0.9) \times 17,550 \quad (\text{Eq. 12.6}) \text{ or} \\ &= z_{0.9} \times \sigma_L = 1.28 \times 17,550 \\ &\approx \mathbf{22,491} \text{ tablets}\end{aligned}$$



12.4. Impact of Supply Uncertainty on Safety Inventory

- Required safety inventory as a function of lead time variability
 - Given CSL = 0.9:

s_L	σ_L	ss (units)	ss (days)
6	15,058	19,298	7.72
5	12,570	16,109	6.44
4	10,087	12,927	5.17
3	7,616	9,760	3.90
2	5,172	6,628	2.65
1	2,828	3,625	1.45
0	1,323	1,695	0.68



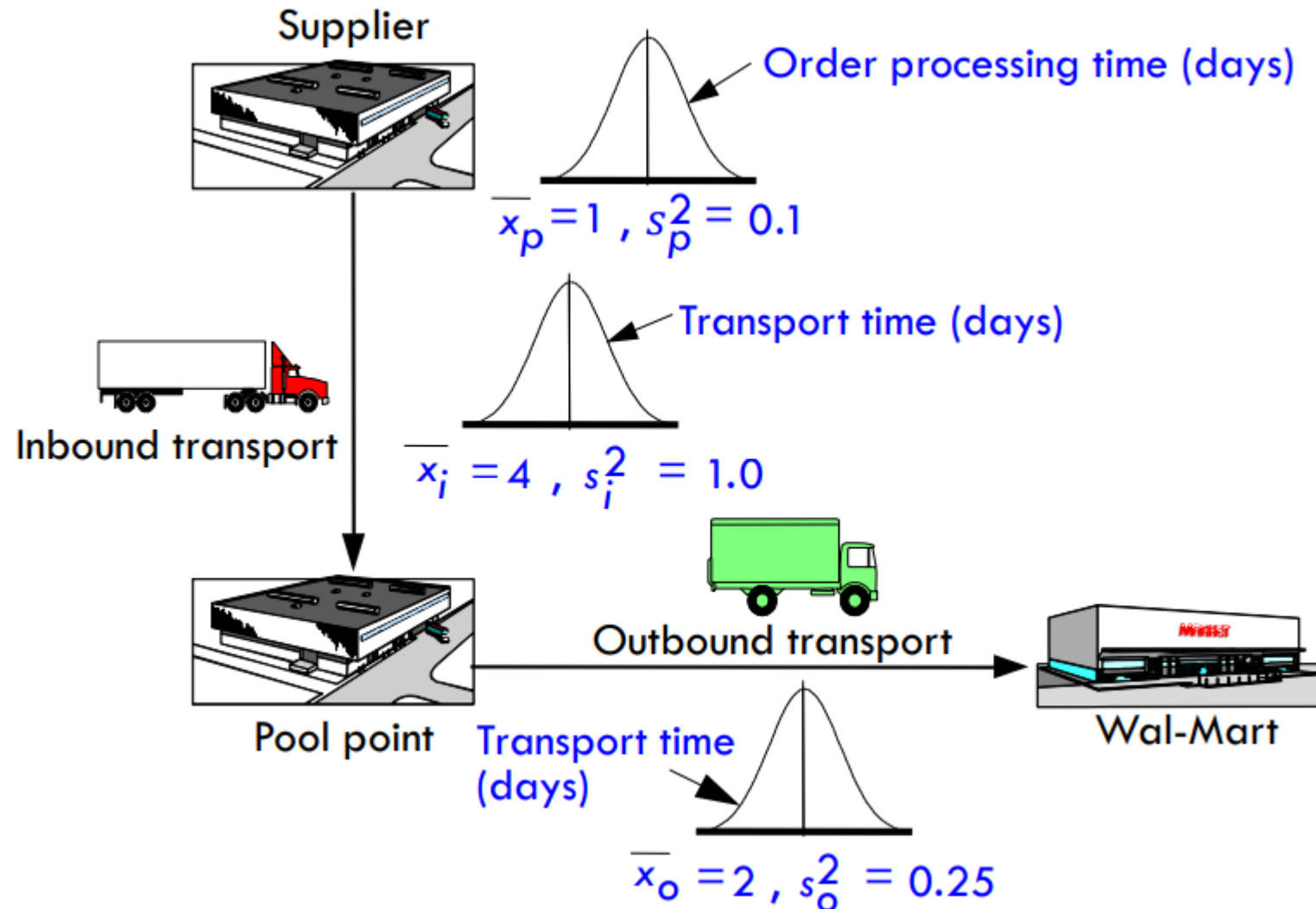
12.4. Impact of Supply Uncertainty on Safety Inventory

- Example: Supply chain with pool point
 - Wal-Mart: Inventory management for picture frames
 - Suppose that inventory is to be maintained on Wal-Mart's shelf
 - Its daily demand for an item (D) $\sim$ NORM(100, 10)
 - Inventory holding cost (h) = 0.1 per year
 - Product cost (C) = \$5/unit, ordering cost (S) = \$10 per order
 - $CSL = 0.99$ ($z_{0.99} = 2.33$)
 - Determine the order quantity, reorder point, and average inventory to be held at Wal-Mart under continuous review policy



12.4. Impact of Supply Uncertainty on Safety Inventory

- Example: Supply chain with pool point



12.4. Impact of Supply Uncertainty on Safety Inventory

- Example: Supply chain with pool point

- Analysis:

Consider the lead-time for the entire channel

$$L = \bar{x}_p + \bar{x}_i + \bar{x}_o = 1 + 4 + 2 = 7 \text{ days}$$

$$s_L^2 = s_p^2 + s_i^2 + s_o^2 = 0.1 + 1.0 + 0.25 = 1.35 \text{ days}$$

Using $\sigma_L = \sqrt{L\sigma_D^2 + D^2s_L^2}$,

$$\sigma_L = \sqrt{7(10^2) + 100^2(1.35)} = \sqrt{14200} = 119 \text{ units}$$

Thus,

$$Q^* = \sqrt{\frac{2(100)(365)(10)}{0.1(5)}} = 1,208 \text{ units}; ss = z_{0.99}(\sigma_L) = 2.33(119) = 278$$

$$ROP = D(L) + ss = 100(7) + 278 = 978 \text{ units}$$

$$AIL = \frac{Q^*}{2} + ss = \frac{1208}{2} + 278 = 882 \text{ units}$$

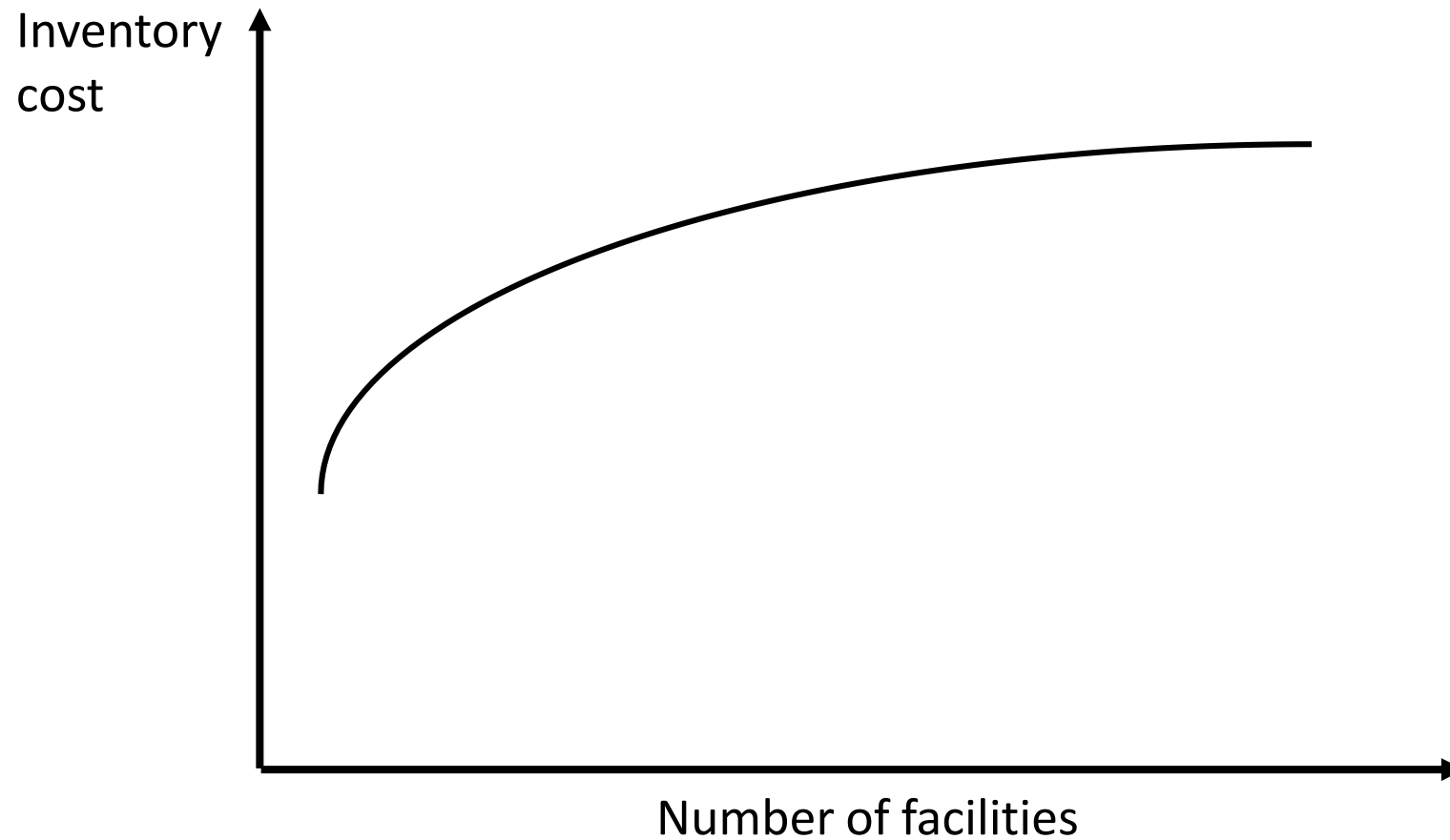


12.4. Impact of Supply Uncertainty on Safety Inventory

- Reasons of supply lead time variability
 - Suppliers' poor production planning
 - Buyers' behavior
 - Same day orders of multiple items to a supplier or all suppliers, and possibly same day arrivals of multiple deliveries
 - Frequent change of order quantity and delivery date
 - Transportation
 - Lack of capacity, unavailability, poor scheduling



12.5. Impact of Aggregation on Safety Inventory



12.5. Impact of Aggregation on Safety Inventory

- Impact of aggregation on safety inventory
 - Notations
 - D_i : mean weekly demand in region i , $i = 1, \dots, k$
 - σ_i : standard deviation of weekly demand in region i , $i = 1, \dots, k$
 - k : number of regions
 - ρ_{ij} : correlation of weekly demand for regions i and j , $1 \leq i \neq j \leq k$
 - σ_D^C : standard deviation of weekly demand when inventories are centralized (aggregated) into one region
 - L : lead time
 - CSL : cycle service level



12.5. Impact of Aggregation on Safety Inventory

- Impact of aggregation on safety inventory
 - Total safety inventory in decentralized option

$$SS = \sum_{i=1}^k [F_s^{-1}(CSL) \text{ or } z_{CSL} \times (\sqrt{L} \times \sigma_i)]$$

If $\sigma_i = \sigma_D \forall i$,

$$SS = k \times F_s^{-1}(CSL) \text{ or } z_{CSL} \times \sqrt{L} \times \sigma_D$$

- Total safety inventory on aggregation option

$$D^C = \sum_{i=1}^k D_i$$

$$Var(D^C) = \sum_{i=1}^k \sigma_i^2 + 2 \sum_{i>j} \rho_{ij} \sigma_i \sigma_j$$

$$SS = F_s^{-1}(CSL) \text{ or } z_{CSL} \times (\sqrt{L} \times \sigma_D^C) \quad \text{where } \sigma_D^C = \sqrt{Var(D^C)}$$

If all k regions have demand that is independent and identically distributed with D and σ_D , then $D^C = kD$ and $\sigma_D^C = \sqrt{k} \sigma_D$.

Thus,

$$SS = F_s^{-1}(CSL) \text{ or } z_{CSL} \times (\sqrt{L} \times \sqrt{k} \times \sigma_D)$$



12.5. Impact of Aggregation on Safety Inventory

- Impact of aggregation on safety inventory
 - Safety inventory savings on aggregation

$$F_S^{-1}(CSL) \text{ or } z_{CSL} \times \sqrt{L} \times (\sum_{i=1}^k \sigma_i - \sigma_D^C)$$

- The safety inventory savings on aggregation increase with the desired cycle service level (CSL) and replenishment lead time (L)
- The safety inventory savings on aggregation increase as the demand correlation for regions (ρ_{ij}) decrease (or the safety inventory savings on aggregation decrease as the demand correlation for regions increase). Thus, the safety inventory savings on aggregation is the largest when ρ_{ij} are equal to -1.

$$(\text{ref.}) \quad \sigma_D^C = \sqrt{\sum_{i=1}^k \sigma_i^2 + 2 \sum_{i>j} \rho_{ij} \sigma_i \sigma_j}$$



12.5. Impact of Aggregation on Safety Inventory

- Example 12.8: Impact of correlation on value of aggregation
 - BMW Car Dealer: Four Dealership Locations (**Disaggregated**)
 $D = 25$ cars/week, $\sigma_D = 5$ cars/week, $L = 2$ weeks, $CSL = 0.9$
 - What would the effect be on safety stock if the four outlets are consolidated into one large outlet (aggregated)?
 - **Analysis:**

At each disaggregated outlet:

For $L = 2$ weeks, $\sigma_L = \sqrt{2} \times 5 = 7.07$ cars

$ss = \text{NORMSINV}(0.9)$ or $z_{0.9} \times 7.07 = 9.06$

Each outlet must carry nine cars as safety stock inventory
so, safety inventory for the four outlets in total is $(4)(9) = 36$ cars



12.5. Impact of Aggregation on Safety Inventory

- Example 12.8: Impact of correlation on value of aggregation
 - One outlet (i.e., **aggregated option**) when $\rho_{ij} = 0$:

$$D^C = D_1 + D_2 + D_3 + D_4 = 25 + 25 + 25 + 25 = 100 \text{ cars/week}$$

$$\sigma_D^C = \sqrt{4 \times 5^2} = \sqrt{4} \times 5 = 10 \text{ cars/week}$$

$$\sigma_L^C = \sqrt{L} \sigma_D^C = \sqrt{2} (10) = (1.414)(10) = 14.14 \text{ cars}$$

$$ss = \text{NORMSINV}(0.9) \text{ or } z_{0.9} \times 14.14 \cong 18 \text{ cars}$$

- The benefit of aggregation decreases as demand in different areas is more positive correlated



12.5. Impact of Aggregation on Safety Inventory

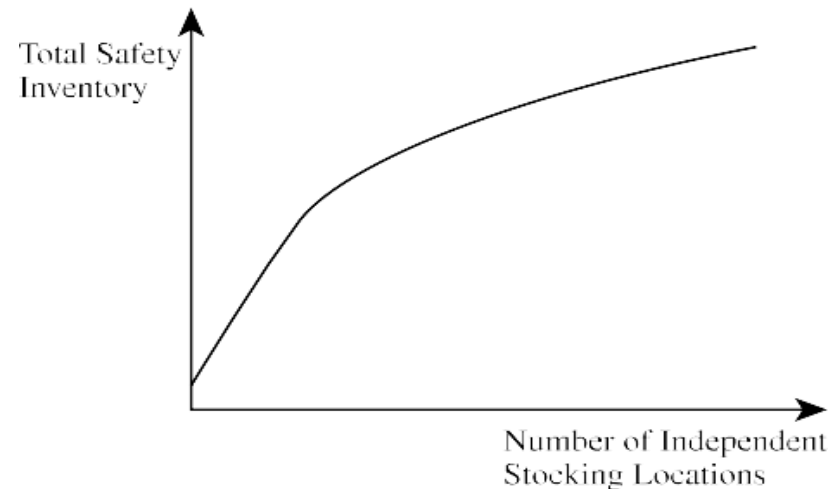
- Safety inventory in the disaggregate and aggregate options

ρ	Disaggregate Safety Inventory	Aggregate Safety Inventory
0.0	36.25	18.12
0.2	36.25	22.93
0.4	36.25	26.88
0.6	36.25	30.33
0.8	36.25	33.42
1.0	36.25	36.25



12.5. Impact of Aggregation on Safety Inventory

- Risk pooling
 - **Risk pooling**: Concept that the variability (or uncertainty) in demand is reduced by aggregating demand across several regions
 - Offset effect of high- and low-demand regions, which is increased and thus reduces inventories more **as the demand between them is more negatively correlated** (or as $\rho_{low,high}$ is close to -1)
(e.g.) Most online retailers
 - **Square-root law**: If the number of independent stocking locations with about the same demand size decreases by a factor of n through aggregation, the average safety inventory (also cycle inventory) is expected to decrease by a factor of $\sqrt{n}$



12.5. Impact of Aggregation on Safety Inventory

- Benefit and disadvantages of aggregation
 - **Benefits**
 - Reduction in cycle and safety inventories in the supply chain while maintaining the same level of product availability
 - Improvement of product availability with the same inventory in the supply chain
 - **Disadvantages**
 - Increase in response time to customer order
 - Increase transportation cost to customer



12.5. Impact of Aggregation on Safety Inventory

- Various methods of aggregation
 - Five methods of aggregation without having to physically centralize all inventories in the supply chain
 - Information centralization
 - Specialization
 - Product substitution
 - Component commonality
 - Postponement



12.5. Impact of Aggregation on Safety Inventory

- (1) Information centralization
 - **Virtual aggregation** (or centralization) of inventories carried at multiple locations using information system
 - Customer order is filled from the closest stock locations in case of a stock-out
 - Online information system that allows access to current inventory records in stock locations from each location
 - Reduced safety inventory while providing a high level of product availability
 - Effective especially when the correlations of demand between locations are negative
 - Example: Gap, Wal-Mart, Redbox



12.5. Impact of Aggregation on Safety Inventory

- (2) Specialization
 - Whether all items should be stocked at every location?
 - Products have different demands in different locations
 - (e.g.) A wide variety of snow boots at stores in Texas. **Sensible?**
 - **Specialize (or stock)** in one central location if the reduction in safety inventory is large by aggregation
 - Safety inventory vs. response time and transportation cost
 - Benefits of aggregation by specialization can be affected by:
 - Coefficient of variation ($cv = \sigma/\mu$) of demand
(Higher cv yields greater reduction in safety inventory)
 - Value of item
(High-value items provide more benefits from specialization)



12.5. Impact of Aggregation on Safety Inventory

- (2) Specialization
 - General rules for specialization
 - **Slow-moving items** typically having high cv stock at a centralized location through specialization
 - Because the reduction in safety inventory is important
 - **Fast-moving items** typically having low cv stock at decentralized locations close to the customer
 - Because the improvement in response time and reduction in transportation cost are important



12.5. Impact of Aggregation on Safety Inventory

- (3) Product substitution
 - Product substitution uses of one product to satisfy the demand for another product
 - Two instances
 - **Manufacturer-driven one-way substitution** (e.g., hard drive)
 - The manufacturer substitutes a high-value product for a low-value product that is not in inventory
 - **Customer-driven two-way substitution** (e.g., detergent)
 - Substitution aggregates demand across the products, which reduces safety inventories without hurting product availability
 - The value increases as demand uncertainty increases and demand for two products becomes more negatively correlated
 - The value decreases as the cost differential between products increases in the case of manufacturer-driven one-way substitution



12.5. Impact of Aggregation on Safety Inventory

- (4) Component commonality
 - Using **common components** in a variety of different products is an effective approach to exploit aggregation and reduce safety inventories for components without hurting product availability (e.g., PC, automobiles)
 - As a component is used in more finished products, it needs to be more flexible and complicated, resulting in the increased cost of producing it and possibly causing quality problems resulted from the incomplete compatibility
 - **Trade-off between the increase in component and quality cost and the decrease in safety inventory** when deciding on the appropriate level of component commonality



12.5. Impact of Aggregation on Safety Inventory

- Example 12.11: Value of component commonality
 - Dell: 27 Different PC Models with Three Distinct Components (Hard drive, CPU, RAM)
 - Specific-component option: 3×27 models = 81 distinct components
 - Common-component option: 3×3 model groups = 9 distinct components
 - Monthly demand for each PC model $\sim NORM(5000, 3000)$
 - Replenishment lead time (L) = one month for each component
 - $CSL = 95\%$ for component inventory ($Z_{0.95} = 1.645$)
 - **Analysis:**
 - Total safety stock under specific option:
$$81 \times [NORMSINV(0.95) \text{ or } Z_{0.95}] \times \sqrt{1} \times 3,000 = 399,699 \text{ units}$$
 - Total safety stock under common option:
$$9 \times [NORMSINV(0.95) \text{ or } Z_{0.95}] \times \sqrt{1} \times \sqrt{9} \times 3,000 = 133,233 \text{ units}$$



12.5. Impact of Aggregation on Safety Inventory

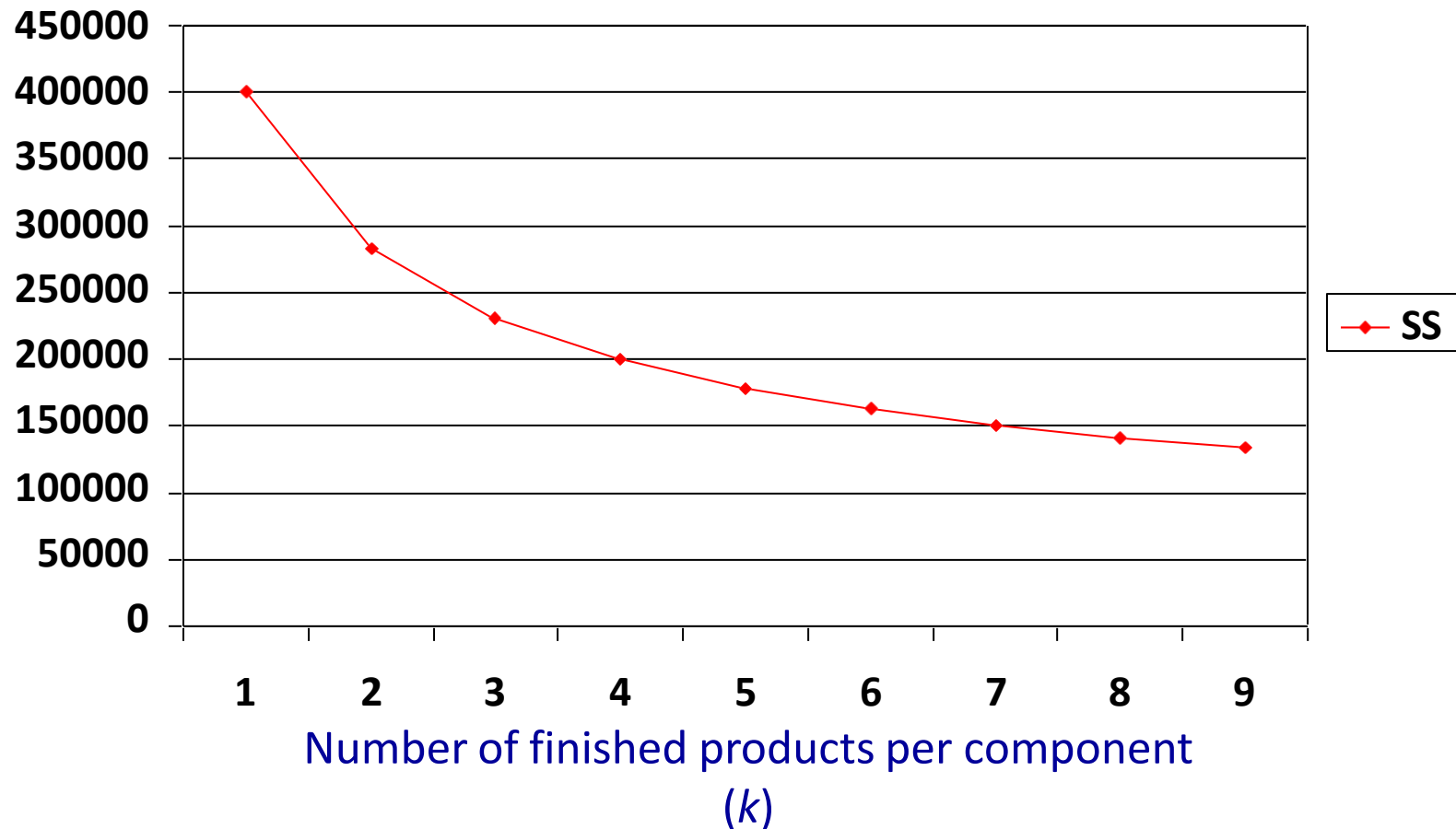
- Marginal benefit of component commodity

Number of Finished Products per Component	Safety Inventory	Marginal Reduction in Safety Inventory	Total Reduction in Safety Inventory
1	399,699		
2	282,630	117,069	117,069
3	230,766	51,864	168,933
4	199,849	30,917	199,850
5	178,751	21,098	220,948
6	163,176	15,575	236,523
7	151,072	12,104	248,627
8	141,315	9,757	258,384
9	133,233	8,082	266,466



12.5. Impact of Aggregation on Safety Inventory

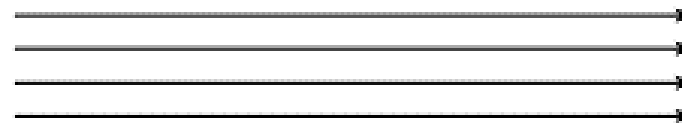
- Benefit of component commodity



12.5. Impact of Aggregation on Safety Inventory

- (5) Postponement

- The ability of a supply chain to delay product differentiation or customization until closer to the time the product is sold (e.g.) Paint, HP printer, Mens trousers



Supply Chain Flows Without Postponement



Supply Chain Flows with Component
Commonality and Postponement

- Classical example: Benetton (colored sweaters)
- Postponement allows a supply chain to exploit aggregation and thus significantly reduce safety inventory, furthermore, to perform accurate demand forecasts and thus improve product availability
- Postponement can be a powerful concept for the e-commerce channel (e.g., PC, furniture)



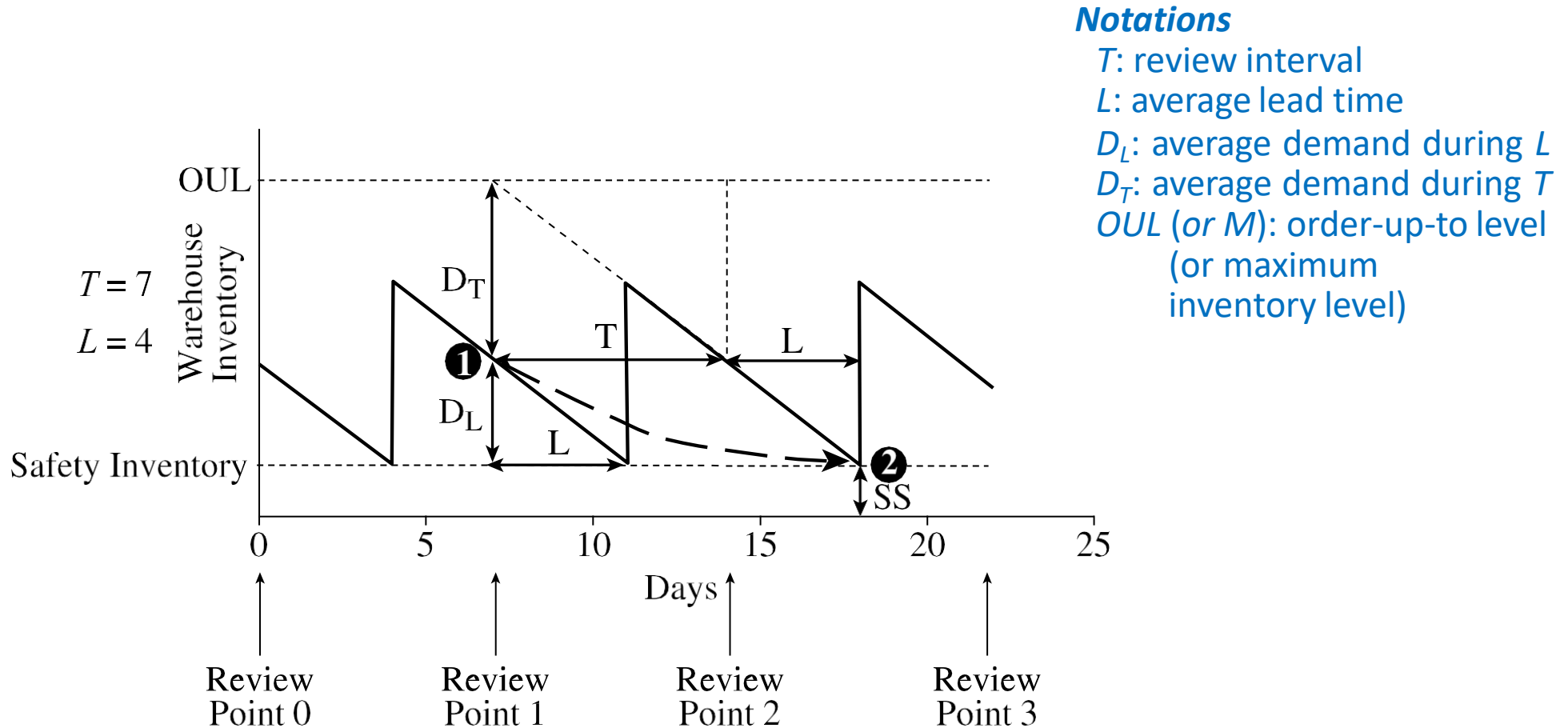
12.6. Impact of Replenishment Policies on Safety Inventory

- Continuous review policy (Q system)
- Periodic review policy (P system)



12.6. Impact of Replenishment Policies on Safety Inventory

- Inventory Profile for Periodic Review Policy with $L = 4$, $T = 7$



12.6. Impact of Replenishment Policies on Safety Inventory

- Periodic Review Policy: Determining Operation Parameters

Given CSL ,

$$SS = F_s^{-1}(CSL) \times \sigma_{T+L} \\ = NORMSINV(CSL) \times \sigma_{T+L} \text{ or}$$

$$SS = Z_{CSL} \times \sigma_{T+L}$$

$$OUL \text{ (or } M) = D_{T+L} + SS$$

where

$$D_{T+L} = (T + L)D$$

$$\sigma_{T+L} = \sigma_D \sqrt{T + L}$$

Notations

σ_D : standard dev. of demand per period

σ_{T+L} : standard dev. of demand for $T + L$

ss : safety inventory

CSL : desired cycle service level

$= Prob(\text{Demand during } T + L \leq OUL \text{ or } M)$

(c.f.) For continuous review policy (Q system):

$$SS = F_s^{-1}(CSL) \text{ or } Z_{CSL} \times \sigma_L$$



12.6. Impact of Replenishment Policies on Safety Inventory

- Example 12.13: Evaluating safety stock for a periodic review system

- Carrefour: Lego Sale

Weekly demand $\sim NORM(2500, 500)$ boxes, $L = 2$ weeks,

$T = 4$ weeks, $CSL = 0.9$

- Analysis:

$$D_{T+L} = 15,000, \sigma_{T+L} = \sqrt{T+L} \sigma_D = 1,225, \sigma_L = \sqrt{L} \sigma_D = 707$$

- Continuous review policy (Q system):

$$ss = F_S^{-1}(CSL) \text{ or } z_{CSL} \times \sigma_L, ROP = D_L + ss$$

- Periodic review policy (P system):

$$ss = F_S^{-1}(CSL) \text{ or } z_{CSL} \times \sigma_{T+L}, OUL(M) = D_{T+L} + ss$$

- Results:

$$\left\{ \begin{array}{l} \text{Q system: } ss = 906; ROP = 5,906 \\ \text{P system: } ss = 1,570; OUL = 16,570 \end{array} \right.$$



12.8. Managerial Levers to Reduce Safety Inventory

- Reduction in supply uncertainty
 - Coordinated demand and supply planning
- Reduction in lead time
 - Reduction in production time, transportation time, and delays
 - Delays (that arise in the processes of order placement, production planning, and transportation planning) contribute more to lead times than the production and transportation time
- Reduction in demand uncertainty
 - Reduce information distortion (i.e., bullwhip effect) through the sharing of the buyer's actual sales information
 - Aggregate demand through physical aggregation, information aggregation, product substitution, component commonality, and postponement

